Gravitational Density:

Massive States, Mimetic Matter, and Vacuum Stress

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Abstract

This paper compares three descriptions of gravitationally relevant density: a dilute population of massive Yang–Mills-inspired states, the matter-like contribution in mimetic gravity, and vacuum stress. The motivating question is whether a positive excitation mass scale, an effective density expressed through gravitational variables, and a vacuum contribution can support a common description of an intrinsic gravitational contribution of space. The comparison is organized by the comoving energy balance $\mathrm{d}(J\rho)/\mathrm{d}t = -p\dot{J} + JQ$, with pressure and exchange retained explicitly.

The calculations identify a conditional correspondence. Conserved massive populations in a pressureless description have density $\sum_i m_i N_i/J$, while mimetic matter has density C/J; matching the coefficient and four-velocity matches their effective dust stress tensors. A two-species finite-temperature calculation quantifies the approach to this description rather than imposing zero pressure. Separately conserved vacuum stress instead has $p = -\rho$ and constant density. Assigning it a matter-like density history requires a nonzero exchange term, calculated and independently integrated here.

The contribution is a comparative exposition: matching density is distinguished from matching stress and from identifying physical origin. Numerical checks verify the implementation of established formulas and illustrate their consequences; they are not independent evidence for an intrinsic density of space. The comparison does not construct a Yang–Mills mass gap. It specifies why the proposed identification of dark matter with space itself requires more than a common mass scale or homogeneous density history.

A focused extension completes a candidate dust-plus-vacuum stress with its gravitational coupling and conservation laws. A linear Fourier mode on two coupled backgrounds connects density evolution to the metric and to a clock-referenced light-travel delay. The mixed example's evolving potential gives a delay approximately 0.775% below its frozen-potential reference. Under matched dynamics and initial data, this candidate reproduces the corresponding dust gravitational response without changing locally measured light speed; that effective agreement does not select the physical origin of its density.

1 Introduction

The author's motivating interpretation is that space itself may possess an intrinsic gravitational contribution, rather than being only a setting containing massive objects. Its possible relationship to dark matter raises a more specific mathematical question: does a positive mass scale in a field theory

determine such a contribution, or are further population, dynamical, and vacuum assumptions indispensable? This paper investigates that question by comparing three established descriptions on common variables before assigning them a common physical origin.

The first description uses massive states inspired by a Yang–Mills spectrum. The second uses an effective matter-like density expressed through the gravitational variables of mimetic gravity. The third is vacuum stress. They concern different objects: an excitation spectrum, a gravitational model with additional dynamical content, and a stress contribution with negative pressure. The formal Yang–Mills mass-gap question concerns a positive spectral separation above the vacuum in a constructed quantum field theory [1, Sec. 4]. A population calculation using supplied masses addresses a consequence of massive states, not that construction.

The contribution here is a source-based analytical synthesis with controlled numerical illustrations. It makes the coefficient matching between cold massive matter and mimetic density explicit, explains why matching density alone does not match vacuum stress, and calculates the exchange required to impose a common density curve despite different pressures. These are consequences of established stress and conservation laws, not new laws or a new microscopic identification. The numerical work checks their implementation and displays the finite-temperature and volume-history examples used in the comparison.

Section 2 sets out the shared variables and energy balance. Sections 3–5 develop the three descriptions and their numerical examples. Section 6 states their conditional correspondence and its implications for the motivating interpretation. Section 7 develops the dust-plus-vacuum option as a specified candidate, including its linear inhomogeneous response and light-travel timing. The conclusion distinguishes the completed mathematical comparison from the stronger physical identification that motivated it.

2 Shared Stress and Conservation Description

A meaningful density comparison must retain pressure, the volume history, and any energy transfer. These quantities determine whether two initially equal densities continue to agree and whether they represent the same stress tensor.

2.1 Conventions and the common integral

Use metric signature $(-+++)$, $c = \hbar = k_B = 1$, and a future-directed unit four-velocity u^μ, so that $u^\mu u_\mu = -1$. For an isotropic rest-frame stress,

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu + p g^{\mu\nu}, \qquad T^\mu{}_\mu = -\rho + 3p. \tag{2.1}$$

Here ρ is energy density, not necessarily rest-mass density. Let t be comoving proper time and $J(t) > 0$ a comoving volume relative to a fixed reference volume, normalized to $J(0) = 1$ in the numerical examples. Let Q denote energy transfer into the selected component per unit proper time and physical volume. The homogeneous energy balance and its integral are

$$\dot\rho + \frac{\dot J}{J}(\rho + p) = Q, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(J\rho) = -p\dot J + JQ, \tag{2.2}$$

$$\rho(t) = \frac{J(0)\rho(0) + \displaystyle\int_0^t [-p(s)\dot J(s) + J(s)Q(s)]\,\mathrm{d}s}{J(t)}. \tag{2.3}$$

Thus the energy in a comoving volume changes through pressure work and exchange. The integral is a common representation, not a closed evolution law until the pressure relation, geometry, transfer, and initial data have been supplied.

For an isotropic scale factor a, $J = a^3$. Where a changes and $\rho > 0$, separate conservation gives

$$\frac{\mathrm{d}\log\rho}{\mathrm{d}\log a} = -3(1+w), \qquad w = \frac{p}{\rho}, \qquad Q = 0. \tag{2.4}$$

Pressureless matter therefore has inverse-volume density; vacuum-form pressure gives constant density. The distinction follows before numerical integration.

2.2 Inputs and comparison domain

Table 2.1 separates the data entering the three descriptions. A mass and an energy density have different dimensions: in the units used here they scale as energy and energy to the fourth power, respectively. Turning a mass scale into a density requires additional information.

Description	Supplied data	Quantity compared
Massive states	Masses, multiplicities, temperature, population parameters	Energy and pressure moments; conserved rest-energy coefficient
Mimetic matter	Gravitational/scalar construction and integration data C	Effective dust density C/J
Vacuum stress	Vacuum density and separate-conservation or exchange assumption	Pressure $-\rho_v$ and corresponding density history

Table 2.1: Distinct inputs retained throughout the comparison. A common stress description does not determine these inputs from one another.

The initial three-way calculations concern isotropic gas snapshots and homogeneous density evolution on prescribed positive volume histories. A flat dust-only reference and a flat vacuum-only reference also satisfy their respective homogeneous gravitational field equations. Other volume histories test conservation alone. No observational dark-matter abundance or microscopic density of space is fitted. Sections 3–6 concern homogeneous comparisons; the additional linear perturbation example is specified separately in Section 7. No formation of gravitationally bound structures is simulated. A homogeneous background does not itself imply vacuum stress: matter and radiation can also have homogeneous backgrounds with different pressure and perturbation dynamics [5].

3 Massive States and the Pressureless Description

The first calculation asks what follows from specifying massive excitations and what still depends on their population. It also quantifies how far the finite-temperature state lies from the pressureless description needed for an exact dust correspondence.

3.1 Mass inputs and gas moments

Forestell, Morrissey, and Sigurdson discuss a Yang–Mills dark sector with a lattice-informed glueball spectrum and a population history governed by reactions and decays [2, Sec. II.A and Sec. IV]. Their quoted $SU(3)$ ratio for the 1^{+-} and 0^{++} states is $m_1/m_0 = 1.71(5)$. Here the central value is held fixed, with masses $(m_0, m_1) = (1, 1.71)$ and spin multiplicities $(d_0, d_1) = (1, 3)$. This is a two-species illustrative truncation, not a calculation of the full spectrum or the source's cosmological reaction network.

For species i, take the dilute Maxwell–Boltzmann distribution

$$f_i(k) = z_i \exp[-E_i(k)/T], \qquad E_i(k) = \sqrt{k^2 + m_i^2}, \qquad x_i = \frac{m_i}{T}, \tag{3.1}$$

where k is momentum magnitude, T temperature, and z_i fugacity.

The number, energy, and pressure moments are

$$n_i = \frac{d_i}{2\pi^2} \int_0^\infty k^2 f_i(k)\, \mathrm{d}k, \tag{3.2}$$

$$\rho_i = \frac{d_i}{2\pi^2} \int_0^\infty k^2 E_i(k) f_i(k)\, \mathrm{d}k, \qquad p_i = \frac{d_i}{6\pi^2} \int_0^\infty \frac{k^4}{E_i(k)} f_i(k)\, \mathrm{d}k. \tag{3.3}$$

Writing K_α for the modified Bessel function of the second kind gives

$$n_i = \frac{d_i z_i m_i^2 T}{2\pi^2} K_2(x_i), \qquad \rho_i = n_i \left[m_i \frac{K_1(x_i)}{K_2(x_i)} + 3T \right], \qquad p_i = n_i T. \tag{3.4}$$

These formulas follow directly from the stated distribution. In the number integral, $k = m_i \sinh\theta$ and the recurrence $K_3 - K_1 = 4K_2/x_i$ yield the first expression. Differentiation with respect to inverse temperature at fixed fugacity yields energy density. Integration by parts yields $p_i = n_i T$, with vanishing endpoint terms.

3.2 Population dependence and finite-temperature results

The calculation uses 41 logarithmically spaced values of m_0/T from 8 to 80 and common fugacities $z_0 = z_1 \in \{0.25, 1, 4\}$, giving 123 combined states. The largest occupancy in the specified distribution is $4e^{-8} \simeq 0.00134$, consistent with its dilute classical treatment. These are separate temperature snapshots at fixed fugacity, not a cooling trajectory with conserved particle number.

With $\rho = \sum_i \rho_i$ and $p = \sum_i p_i$, both moments are linear in the common fugacity. Consequently, at fixed temperature and masses, the density at $z = 4$ is 16 times that at $z = 0.25$ (Figure 3.1). This exact parameter dependence illustrates why a mass scale alone does not set a populated density.

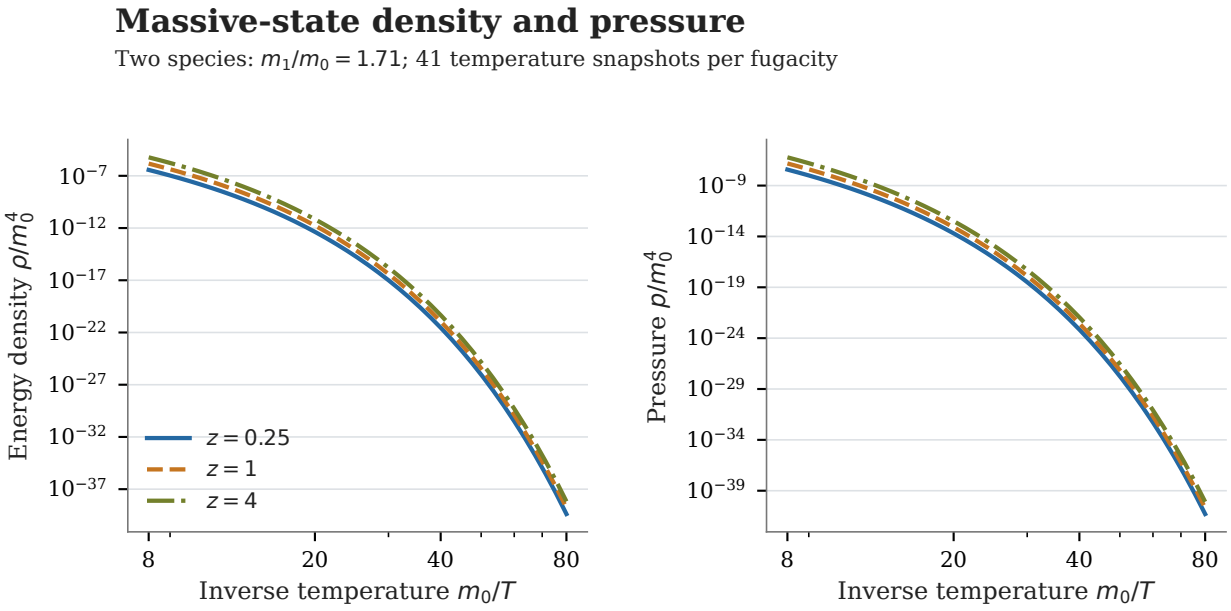


Figure 3.1: Energy density and pressure of the specified two-species population. Mass inputs follow Ref. [2]; the gas moments are evaluated from Eq. (3.4). Curves show temperature snapshots, with logarithmic axes. The population changes while the masses remain fixed.

Define the rest-energy density by $\rho_{\text{rest}} = \sum_i m_i n_i$. The cold expansion gives

$$\frac{\rho_i}{n_i} = m_i + \frac{3}{2}T + O(T^2/m_i), \qquad \frac{p_i}{\rho_i} \longrightarrow 0, \qquad \frac{\rho_i}{m_i n_i} \longrightarrow 1 \quad (T/m_i \longrightarrow 0). \tag{3.5}$$

At $m_0/T = 8$, the combined state has $p/\rho = 0.101809$ and $\rho/\rho_{\text{rest}} = 1.210146$. At $m_0/T = 80$, these become 0.0122665 and 1.019039. Thus pressure is about 1.23% of energy density at the coldest tested state, while kinetic energy contributes about 1.90% above rest energy (Figure 3.2). The ratios approach dust behavior, but no finite-temperature state in the calculation is exactly pressureless.

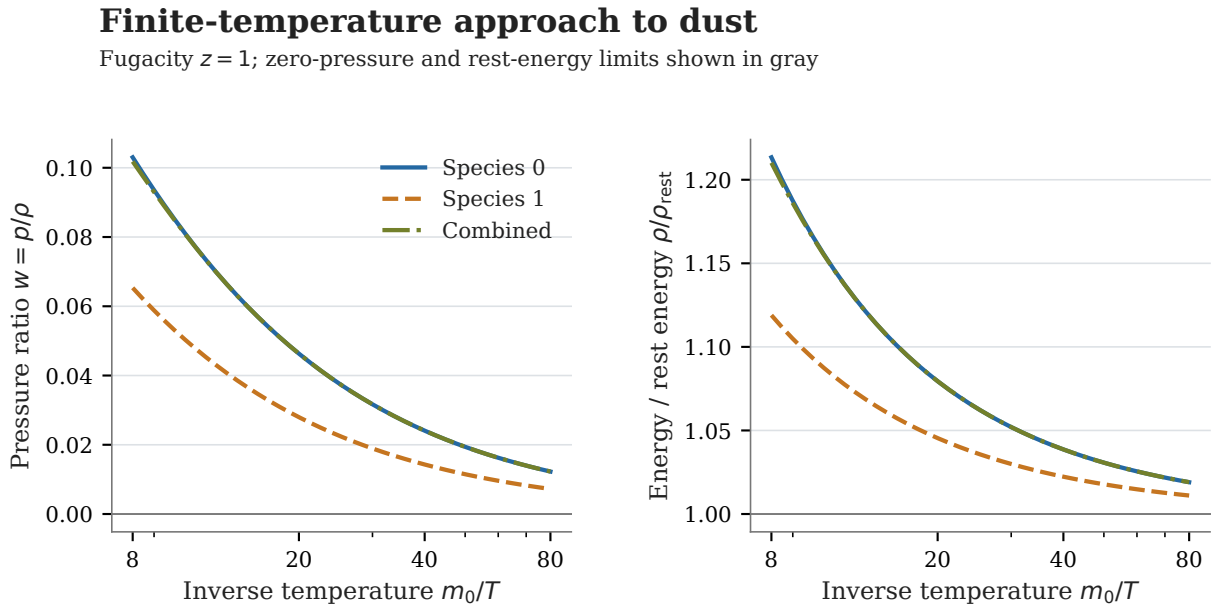


Figure 3.2: Pressure and energy ratios at fugacity 1. The two species and their sum approach the zero-pressure and rest-energy limits. These ratios describe the finite-temperature approximation; a nonzero conserved dust density is a separate effective description, not a claim that fixed-fugacity cooling retains a fixed population.

3.3 Conserved population and numerical verification

For a population transported with a common comoving volume and without number-changing reactions,

$$\frac{\mathrm{d}}{\mathrm{d}t}(Jn_i) = 0, \qquad N_i = Jn_i, \qquad \rho_{\mathrm{rest}}(t) = \frac{\sum_i m_i N_i}{J(t)}. \tag{3.6}$$

Since J is a relative volume, N_i denotes number per reference volume. Equation (3.6) is exact for rest energy under these assumptions. The associated pressureless stress approximation neglects kinetic energy and velocity dispersion; finite-temperature total energy still obeys the pressure-work term in Eq. (2.2). The inverse-cube number law for $J = a^3$ was checked on 101 scale factors from 1 to 10.

To verify the moment evaluation independently, the momentum integrals were transformed using $\eta = (E_i - m_i)/T$ and their common exponential suppression was factored out. Adaptive quadrature was performed at $m_0/T = 8, 20, 80$, for both species and all three moments, with upper cutoffs 40 and 60 in η. These 36 evaluations agreed with exponentially scaled Bessel expressions to a maximum relative difference of 3.82×10^{-15}, against the preselected 10^{-8} comparison tolerance. Quadrature used relative and absolute tolerances 10^{-11} and 10^{-10}, respectively. This is agreement between numerical evaluations, not a rigorous interval enclosure of roundoff or the infinite tail.

The result needed for the comparison is therefore specific: a conserved cold population supplies a matter-like coefficient $\sum_i m_i N_i$, but neither the masses nor the cold limit determine that coefficient. A zero-population control removes excitation density; it makes no determination of gravitational vacuum energy.

4 Matter-Like Density from Gravitational Variables

The second description tests whether the same inverse-volume behavior occurs when an effective density is represented through gravitational variables. It provides the closest direct correspondence to the conserved massive population.

4.1 Dust stress and integration data

Chamseddine and Mukhanov's mimetic construction introduces a physical metric through an auxiliary metric and a scalar field, making an additional conformal degree of freedom dynamical [3, Eqs. (1), (6)–(19)]. Its effective contribution has dust form. After converting the source's $(+---)$ signature to the convention of Eq. (2.1), write

$$\nabla_\mu\phi\nabla^\mu\phi = -1, \qquad T^{\mu\nu}_{\mathrm{mim}} = \rho_{\mathrm{mim}}u^\mu u^\nu, \qquad \nabla_\mu(\rho_{\mathrm{mim}}u^\mu) = 0. \tag{4.1}$$

Here u is the future-directed unit flow associated with the scalar ϕ. In synchronous comoving coordinates, the local spatial volume is proportional to $\sqrt{\det\gamma}$, and conservation integrates to

$$\rho_{\mathrm{mim}}(t,\mathbf{x}) = \frac{C(\mathbf{x})}{J(t,\mathbf{x})}, \qquad p_{\mathrm{mim}} = 0. \tag{4.2}$$

The source's unnormalized volume convention is absorbed into C. Physical density is unchanged by the signature conversion, while the mixed stress trace is $-\rho_{\mathrm{mim}}$ in our convention. This model contains additional dynamical content; it is not ordinary general relativity with an automatically nonzero density of empty space.

The homogeneous examples take constant nonnegative C, including a zero control. The inverse-volume relation holds where the comoving construction and positive volume are defined. Its integration data are supplied, just as the populations were supplied in Section 3. This effective dust correspondence does not establish the model's behavior through caustics, where the single-flow description can fail; that issue is explicitly analyzed in the mimetic literature [6].

4.2 Volume histories and a coupled dust reference

Let $\tau = H_*t$, $r = \rho_{\mathrm{mim}}/\rho_*$, and $r(0) = 1$. Table 4.1 gives the four positive volume histories on $0 \le \tau \le 8$. The density equation is

$$r' = -\frac{J'}{J}r, \qquad r = J^{-1}, \tag{4.3}$$

where a prime denotes differentiation with respect to τ.

Volume history	$J(\tau)$	$r(8)$
Power law	$(1+\tau)^{3/2}$	$1/27$
Dust reference	$(1+3\tau/2)^2$	$1/169$
Exponential	$\exp(3\tau/4)$	e^{-6}
Anisotropic-volume example	$1+\tau$	$1/9$

Table 4.1: Prescribed volumes used for conservation checks. Only the dust reference is also checked here as a coupled matter-only homogeneous gravitational solution. The anisotropic example specifies a volume determinant, not a uniquely determined metric.

For the flat dust-only reference, choose $H_*^2 = 8\pi G\rho_*/3$ and no vacuum contribution. Then

$$a = (1 + 3\tau/2)^{2/3}, \qquad h = \frac{a'}{a} = \frac{1}{1 + 3\tau/2}, \qquad r = a^{-3}, \tag{4.4}$$

and both normalized homogeneous field equations hold:

$$h^2 = r, \qquad \frac{a''}{a} = -\frac{1}{2}r. \tag{4.5}$$

This additional check distinguishes conservation on a chosen geometry from a geometry and density solving the field equations together.

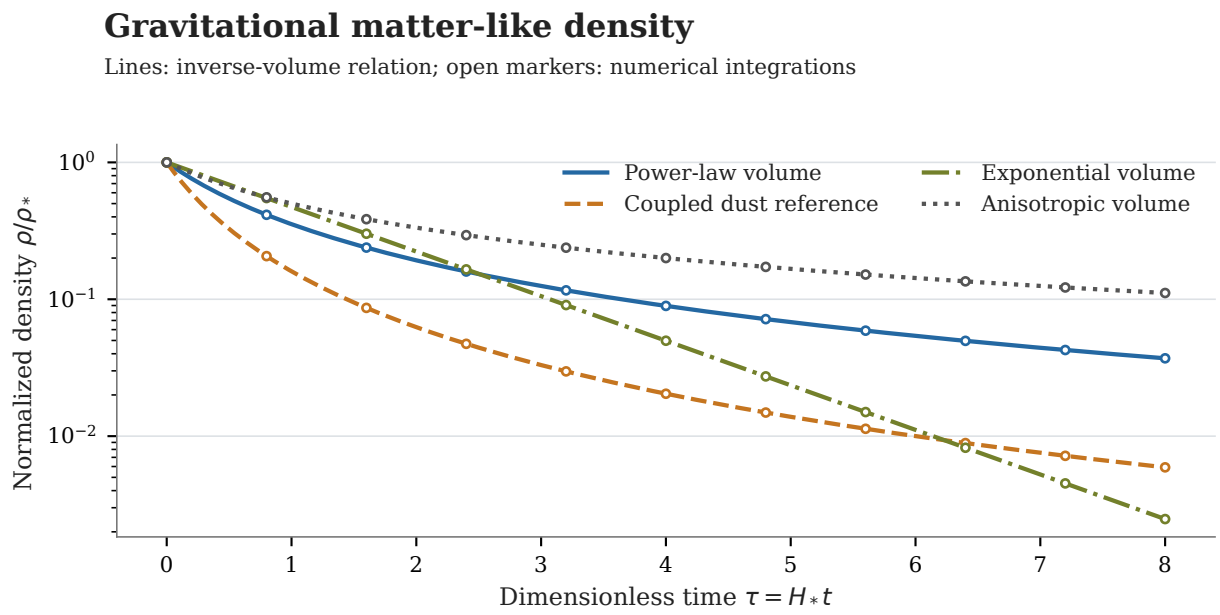


Figure 4.1: Inverse-volume density and independent numerical integration of Eq. (4.3). The effective dust description follows Ref. [3]. All histories begin at normalized density 1; their differing endpoint densities reflect their prescribed volume growth. The curves are illustrative calculations, not observed cosmological histories.

4.3 Numerical method and significance

Each nonzero history was integrated by an adaptive explicit Runge–Kutta method of orders 5 and 4. The baseline used relative tolerance 10^{-9}, absolute tolerance 10^{-12}, and 401 output times; refinement used 10^{-10}, 10^{-13}, and 801 output times. The maximum step was 0.1 in τ. Four paired histories and a zero-density control gave nine integrations.

The maximum relative density discrepancy was 3.92×10^{-9} at baseline and 3.99×10^{-10} under refinement, below the 10^{-7} trajectory comparison threshold. Every paired history improved under the tighter settings. Tensor checks in rest and boosted frames verified the dust trace and unit-velocity conventions; the two dust-reference field-equation residuals were at floating-point precision.

The analytical relation supplies the gravitational-variable counterpart to Eq. (3.6): both descriptions carry a conserved density coefficient divided by comoving volume. Numerical agreement verifies

this implementation, rather than independently establishing a common physical origin for the coefficients.

5 Vacuum Stress, Pressure Work, and Required Exchange

Vacuum stress provides a third gravitational contribution, but one with a different pressure relation. This section identifies the consequence of that difference and calculates the exchange needed to assign vacuum-form stress a matter-like density history.

5.1 Separate conservation and density comparison

In the chosen signature the vacuum contribution is [4, Sec. III]

$$T_v^{\mu\nu} = -\rho_v g^{\mu\nu}, \qquad p_v = -\rho_v. \tag{5.1}$$

Metric compatibility gives $\nabla_\mu T_v^{\mu\nu} = -\nabla^\nu \rho_v$. Separate conservation therefore requires constant ρ_v. In the homogeneous balance,

$$\dot{\rho}_v = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(J\rho_v) = \rho_v \dot{J} = -p_v \dot{J}. \tag{5.2}$$

The changing energy in the chosen volume is exactly balanced by negative-pressure work.

On the dust-reference volume, independently assigning equal initial densities to dust and vacuum gives

$$r_{\mathrm{dust}} = (1 + 3\tau/2)^{-2}, \qquad r_v = 1. \tag{5.3}$$

At $\tau = 8$, these are $1/169$ and 1, respectively. Their pressures also differ at the initial time, when their densities agree (Figure 5.1). This comparison places separately conserved components on one prescribed volume; it does not assert that the dust-only scale factor solves a coupled dust-plus-vacuum model.

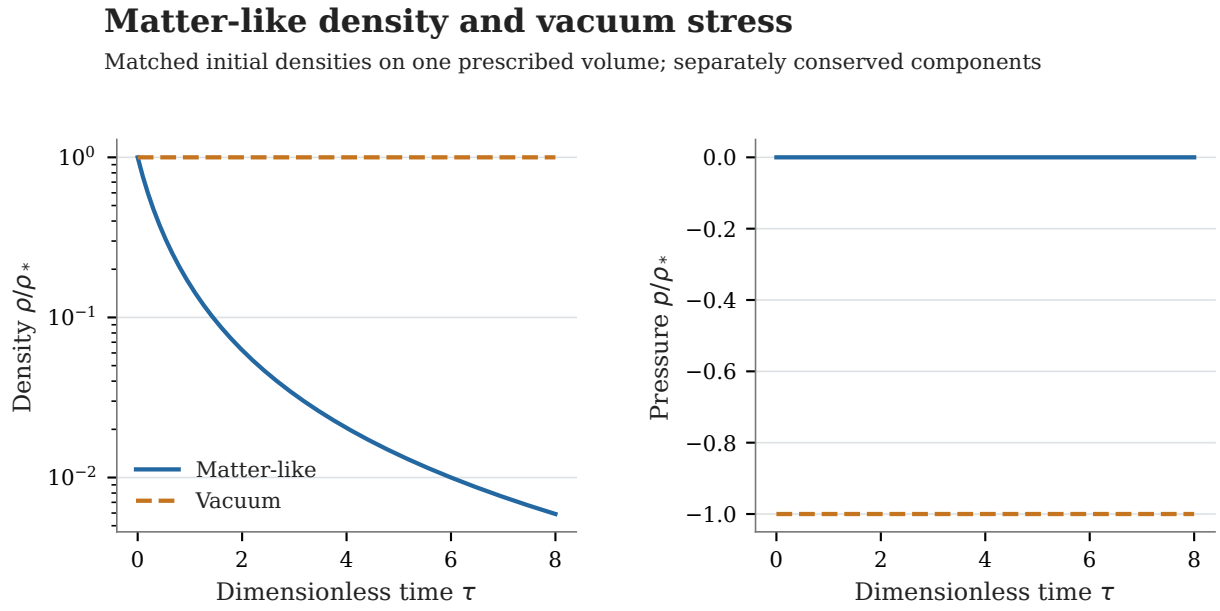


Figure 5.1: Density and pressure for matter-like and vacuum-form contributions on the same volume history. The stress forms are those of Refs. [3, 4]. Equal initial density does not imply equal stress or equal subsequent density evolution.

As a separate coupled reference, a flat positive-vacuum-only solution in the normalization $H_*^2 = 8\pi G\rho_*/3$ has $a = e^\tau$, $r = 1$, $h^2 = r$, and $a''/a = r$. Constant-density integrations also used initial values 0, 1, and 2 on three prescribed backgrounds. Their zero density residuals reflect exact cancellation of $\rho_v + p_v$ in the implemented equation, not observational precision.

5.2 A prescribed matter-like vacuum density

Now retain the vacuum-form pressure but impose the target density $\rho_v = \rho_* a^{-3}$ on the dust reference. Equation (2.2) no longer permits separate conservation. Instead it determines

$$Q = \dot{\rho}_v = -3H\rho_v, \qquad q(\tau) := \frac{Q}{H_*\rho_*} = -\frac{3}{(1+3\tau/2)^3}. \tag{5.4}$$

The negative sign specifies energy transfer out of this component. Integration of $r_v' = q$ from $r_v(0) = 1$ recovers the target density. Figure 5.2 displays both the target and the calculated trajectory, together with the signed exchange.

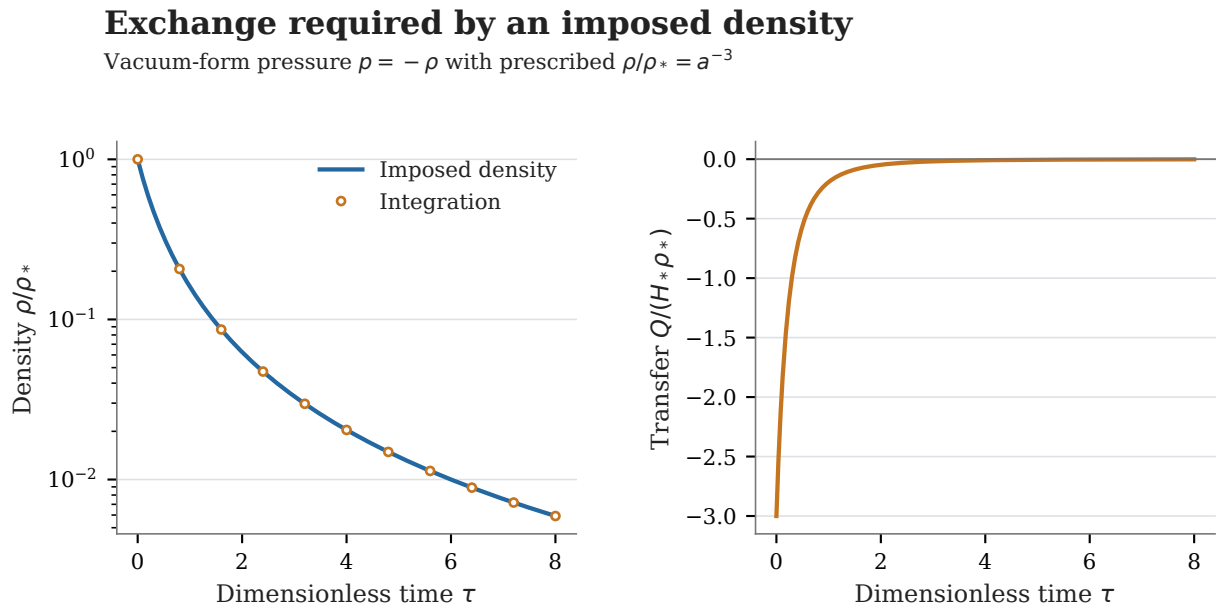


Figure 5.2: Exchange diagnostic for Eq. (5.4). The numerical integration follows the imposed inverse-volume density while retaining vacuum-form pressure. The transfer is derived from the target history, not fitted to observations or inferred as an independently established interaction.

Using the same baseline and refined integration settings as Section 4, the maximum relative discrepancy from the target was 4.82×10^{-8} and 4.93×10^{-9}, respectively. Both met the 10^{-7} threshold. These two integrations and the nine constant-density cases give eleven vacuum-sector integrations. The transfer starts at $q = -3$ and reaches approximately -0.00136550 at $\tau = 8$.

A complete interacting model would also specify the receiving component and conserve total stress. Moreover, in the homogeneous acceleration equation $\ddot{a}/a = -(4\pi G/3)(\rho_{\text{total}} + 3p_{\text{total}})$, with vacuum terms included in the total stress, this component contributes $\rho_v + 3p_v = -2\rho_v$, not the $+\rho_v$ of dust. For positive density its contribution favors acceleration even when exchange makes it dilute as a^{-3}. The total acceleration depends on all components. Thus the calculated exchange matches a density history, not the gravitational behavior of dark matter; it is a conservation diagnostic rather than a proposed dark-matter model.

5.3 Excitation gaps and vacuum reference energy

A positive excitation gap and a vacuum energy density are not interchangeable specifications. For a spectrum measured relative to its vacuum level, a common shift A leaves every difference unchanged:

$$(E_n + A) - (E_0 + A) = E_n - E_0. \tag{5.5}$$

This algebra explains why an excitation gap alone does not determine the reference energy. Once gravity is included, the vacuum stress must still be specified; Eq. (5.5) does not make a gravitational vacuum contribution irrelevant [4]. The calculation therefore retains vacuum density as separate input rather than identifying it with a glueball mass or a population-dependent density.

6 Conditional Correspondence and Physical Interpretation

The three calculations can now be joined without removing their distinct assumptions. The correspondence is exact between effective pressureless descriptions; the numerical gas calculation quantifies a finite-temperature approach to one of them.

6.1 Matching the effective matter contribution

Proposition 6.1 (Conditional stress correspondence)**.** *Consider a common positive comoving volume J and unit four-velocity u. Let constant masses m_i and conserved populations per reference volume N_i define a pressureless rest-energy description. A mimetic dust contribution with integration coefficient C has the same stress tensor exactly when*

$$C = \sum_i m_i N_i. \tag{6.1}$$

A nonzero vacuum-form stress tensor cannot equal this dust tensor merely by matching its energy density in the same rest frame.

Proof. The respective dust tensors are

$$T^{\mu\nu}_{\text{rest}} = \frac{\sum_i m_i N_i}{J} u^\mu u^\nu, \qquad T^{\mu\nu}_{\text{mim}} = \frac{C}{J} u^\mu u^\nu. \tag{6.2}$$

Contracting with $u_\mu u_\nu$ proves necessity of Eq. (6.1); substitution proves sufficiency. For the vacuum comparison, contract instead with the rest-space projector $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$. Dust has $h_{\mu\nu}T^{\mu\nu} = 0$, whereas vacuum stress has $h_{\mu\nu}T_v^{\mu\nu} = -3\rho_v$. Equality requires $\rho_v = 0$, and then equality of the full tensors requires zero dust density as well. □

This is an elementary consequence of the specified stress forms, made explicit for the present comparison. It does not derive C from a Yang–Mills field or identify the additional mimetic degree of freedom with a glueball population. At the tested finite temperatures, the nonzero pressure in Figure 3.2 prevents exact identification of the full gas stress with dust.

6.2 Including vacuum in the same conservation framework

The positive comparison is not limited to choosing one component over another. A common effective dust-plus-vacuum decomposition is

$$T^{\mu\nu} = \frac{C}{J} u^\mu u^\nu - \rho_v g^{\mu\nu}, \qquad \rho_{\text{total}} = \frac{C}{J} + \rho_v, \qquad p_{\text{total}} = -\rho_v. \tag{6.3}$$

For constant C and separately conserved constant ρ_v,

$$\frac{\mathrm{d}}{\mathrm{d}t}(J\rho_{\text{total}}) = \rho_v \dot{J} = -p_{\text{total}}\dot{J}. \tag{6.4}$$

Thus both components fit Eq. (2.3). If the matter coefficient has been matched by Eq. (6.1), particle and mimetic descriptions are alternatives for that coefficient, not two densities to add a second time. Equation (6.3) is an algebraic synthesis at this stage; Section 7 supplies a self-consistent mixed background and its specified linear perturbation.

Comparison	Consequence made explicit	Not determined by that consequence
Massive population	Specified moments, cold ratios, and conserved rest-energy coefficient	Population from the mass gap alone
Mimetic correspondence	Equal dust stress when C, population coefficient, and velocity match	Shared microscopic origin of the matched coefficients
Vacuum comparison	Distinct pressure and constant separately conserved density	Vacuum density from an excitation mass scale
Imposed common density	Required transfer $Q = -3H\rho_v$ for the selected history	Receiving sector or a physical transfer mechanism

Table 6.1: Consequences of the specified established models and the identifications they leave open. Numerical examples illustrate these relations; the table does not claim new microscopic physics.

6.3 What the comparison says about intrinsic density

The motivating interpretation receives a concrete mathematical comparison: a gravitational-variable model admits a matter-like contribution, and its effective stress can match that of conserved cold massive states. This identifies a place where the two descriptions coincide. It also identifies what a stronger interpretation must explain: why the population-dependent coefficient and the mimetic integration coefficient should agree physically, and how any vacuum contribution acquires its pressure and reference density.

Nonzero gravitational density alone is therefore insufficient to identify these descriptions. Their common conservation law accommodates both matter-like and vacuum-like terms, but does not choose their relative amounts or a shared origin. The present calculations neither establish dark matter as space itself nor refute every version of that interpretation. They narrow its mathematical content to specified coefficient, pressure, and exchange relations.

Clustering is a further distinction not evaluated by the homogeneous comparisons above. Cold particle matter and mimetic dust both admit inhomogeneous descriptions; the perturbation evolution, pressure response, and breakdown of a single-flow approximation must be specified before comparing their structure formation [5, 6]. Matching the homogeneous coefficient alone does not establish matching lensing or clustering, nor does it identify which microscopic or gravitational description produces that coefficient. Section 7 supplies those perturbation assumptions for one linear example, without claiming a nonlinear structure-formation calculation.

The numerical work verifies moment formulas, conservation histories, and selected homogeneous reference solutions, with refinement where integration error is relevant. It does not add independent physical evidence to those analytical relations. The result is a conditional effective correspondence and a comparison of its limits, rather than a completed physical unification.

7 A Density Realization, Its Metric Response, and Light Travel

The preceding comparison identifies an effective stress that can now be tested as a candidate realization of the intrinsic-density interpretation. This section completes the dust-plus-vacuum description in Eq. (6.3), follows one linear density mode through the gravitational equations, and calculates its effect on light travel. The purpose is to specify a possible realization and its consequences before adopting it as an account of space itself.

7.1 Two distinctions guiding the realization

First, a dust stress tensor does not identify its physical origin or uniquely select mimetic gravity. The mimetic construction is an established gravitational realization with additional structure [3]; using its effective stress does not itself establish that dark matter is intrinsic to space.

Second, a background equation-of-state parameter $w_s = p_s/\rho_s$ does not alone specify the pressure response of perturbations or anisotropic stress. These affect clustering independently of the homogeneous density history [5, Eq. (30)]. Lim, Sawicki, and Vikman, for example, construct constrained-field models with nonzero pressure but zero sound speed [8]. The candidate below therefore fixes both the background components and their perturbation laws; it does not introduce an unspecified w_s to be selected afterward.

7.2 Stress, conservation, and a possible gravitational realization

Retain $(-+++)$ signature and $c = 1$. Let $\rho_d > 0$ denote the candidate's dust-like density and $\rho_v \geq 0$ a separately conserved constant vacuum density. The subscript s labels the proposed effective contribution, without deciding its underlying origin:

$$T_s^{\mu\nu} = \rho_d u^\mu u^\nu - \rho_v g^{\mu\nu}, \qquad G^{\mu\nu} = 8\pi G\big(T_{\text{other}}^{\mu\nu} + T_s^{\mu\nu}\big), \qquad u^\mu u_\mu = -1. \tag{7.1}$$

Here $G^{\mu\nu}$ is the Einstein tensor. The vacuum contribution is included on the right, so no second copy of it is added as a cosmological constant. Likewise, a contribution already attributed to dark matter is represented once, not counted again under a new interpretation.

Assume $\nabla_\mu T_{\text{other}}^{\mu\nu} = 0$. The contracted Bianchi identity implies $\nabla_\mu T_s^{\mu\nu} = 0$. Expanding this equation and projecting along and perpendicular to u gives

$$u^\mu \nabla_\mu \rho_d + \rho_d \nabla_\mu u^\mu = 0, \qquad u^\mu \nabla_\mu u^\nu = 0, \qquad \rho_d = \frac{C(\mathbf{X})}{J}. \tag{7.2}$$

The last expression holds along a regular comoving flow labeled by $\mathbf{X}$, with $\dot{J}/J = \nabla_\mu u^\mu$. Thus both density transport and geodesic motion, not density transport alone, complete the dust dynamics. The integration function and geometric initial data remain inputs subject to the Einstein constraint equations.

A concrete potential-flow realization is supplied by the established constrained-scalar description of mimetic matter [7, Sec. 2]. In the present conventions it can be written

$$\mathcal{A} = \int \sqrt{-g}\left[\frac{R}{16\pi G} - \frac{\rho_d}{2}\left(g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + 1\right) - \rho_v\right] \mathrm{d}^4x + \mathcal{A}_{\text{other}}. \tag{7.3}$$

Variation of the multiplier ρ_d imposes the unit-gradient constraint. Taking $u_\mu = -\partial_\mu\phi$ future-directed, scalar variation gives continuity and metric variation gives Eq. (7.1) on the constraint. The

normalized gradient also gives geodesic, irrotational flow. This is a candidate realization drawn from existing theory, not a new action claimed here. Its single-flow description has a caustic limitation [6]; the following linear example stays in its small-perturbation regime.

7.3 A self-consistent dust-plus-vacuum background

Set $T_{\text{other}} = 0$ for the illustrative calculation and use a spatially flat Friedmann–Lemaître–Robertson–Walker background. Normalize $a = 1$ at a reference epoch and define $H_* = H(1)$, $\Omega_{d*} = 8\pi G\rho_d(1)/(3H_*^2)$, and $\Omega_{v*} = 1 - \Omega_{d*}$. The coupled equations are

$$E(a)^2 := \frac{H(a)^2}{H_*^2} = \Omega_{d*}a^{-3} + \Omega_{v*}, \qquad \dot{H} = -4\pi G\bar{\rho}_d, \tag{7.4}$$

$$\bar{\rho}_d = \frac{3H_*^2\Omega_{d*}}{8\pi G a^3}, \qquad \rho_v = \frac{3H_*^2\Omega_{v*}}{8\pi G}. \tag{7.5}$$

An overbar denotes the homogeneous dust density and a dot here denotes cosmic proper time. These equations satisfy $\ddot{a}/a = -(4\pi G/3)(\bar{\rho}_d - 2\rho_v)$. The effective background ratio is consequently $w_s = -\rho_v/(\bar{\rho}_d + \rho_v)$; its evolution follows from the fixed components, rather than being a separate fitted parameter. A constant mimetic potential provides the corresponding dust-plus-vacuum behavior in the cited construction [7, Eqs. (24)–(25)].

Two backgrounds are used: $(\Omega_{d*}, \Omega_{v*}) = (1, 0)$ and $(0.3, 0.7)$. They are supplied dimensionless examples, not an observational fit. No radiation, baryons, finite-temperature pressure, or exchange is included in this additional calculation. Unlike a prescribed volume-only check, the density and expansion here satisfy the gravitational equations together.

7.4 One inhomogeneous mode and the metric it produces

Use conformal time η, defined by $\mathrm{d}t = a\,\mathrm{d}\eta$, and conformal Newtonian gauge for scalar perturbations:

$$\mathrm{d}s^2 = a^2\left[-(1+2\Psi)\mathrm{d}\eta^2 + (1-2\Phi)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\right], \qquad \Psi = \Phi. \tag{7.6}$$

The dimensionless potentials are equal because the candidate has no anisotropic stress at linear order. Dust has zero pressure perturbation and $\delta\rho_v = 0$. Define $\delta = \delta\rho_d/\bar{\rho}_d$ in this gauge, the Fourier velocity divergence θ, and $\mathcal{H} = aH$. For a nonzero comoving wavenumber k, the linear conservation and Einstein constraints give [5, Eqs. (23), (43)]

$$\delta' = -\theta + 3\Phi', \qquad \theta' = -\mathcal{H}\theta + k^2\Phi, \tag{7.7}$$

$$k^2(\Phi' + \mathcal{H}\Phi) = 4\pi G a^2\bar{\rho}_d\theta, \qquad k^2\Phi + 3\mathcal{H}(\Phi' + \mathcal{H}\Phi) = -4\pi G a^2\bar{\rho}_d\delta. \tag{7.8}$$

Primes in this subsection denote conformal-time derivatives. Combining the constraints defines the comoving density contrast and its exact linear potential relation:

$$\Delta = \delta + \frac{3\mathcal{H}\theta}{k^2}, \qquad k^2\Phi = -4\pi G a^2\bar{\rho}_d\Delta. \tag{7.9}$$

The distinction between Δ and δ avoids using the short-wavelength Poisson approximation as if it were the full relativistic constraint.

The zero-pressure spatial Einstein equation is $\Phi'' + 3\mathcal{H}\Phi' + (2\mathcal{H}' + \mathcal{H}^2)\Phi = 0$. Substitution of Eq. (7.9), with $q = \log a$, yields

$$D_{qq} + \left(2 - \frac{3}{2}\Omega_d(a)\right) D_q - \frac{3}{2}\Omega_d(a) D = 0, \quad \Omega_d(a) = \frac{\Omega_{d*} a^{-3}}{E(a)^2}, \quad \Delta(a) = \Delta_i \frac{D(a)}{D(a_i)}. \tag{7.10}$$

Here subscripts q denote derivatives with respect to q. The growing mode is fixed by the established integral [9, Eq. (1) and its footnote]

$$D(a) = \frac{5\Omega_{d*}}{2} E(a) \int_0^a \frac{\mathrm{d}b}{b^3 E(b)^3}, \qquad \frac{D(a)}{a} \longrightarrow 1 \quad (a \longrightarrow 0). \tag{7.11}$$

Differentiating gives $D_q = -3\Omega_d(a)D/2 + 5\Omega_{d*}/[2a^2 E(a)^2]$, which fixes the second initial datum without fitting it to the evolved result.

The calculation takes $a_i = 0.05$, $\Delta_i = 10^{-3}$, $k/H_* = 100$, and $a \leq 1.1$, in units $c = 1$. The real spatial mode is $\Delta(a, x) = \Delta(a)\cos(kx)$, with zero mean over a period. Its potential is

$$\Phi(a, x) = \Phi_k(a)\cos(kx), \qquad \Phi_k(a) = -\frac{3\Omega_{d*}}{2(k/H_*)^2 a}\Delta(a). \tag{7.12}$$

Initial velocity and Newtonian-gauge density are fixed by both constraints: $\theta_i = -\mathcal{H}_i \Delta_i D_q(a_i)/D(a_i)$ and $\delta_i = \Delta_i - 3\mathcal{H}_i\theta_i/k^2$. The density contrast remains below 0.02201 in magnitude and $|\Phi| \leq 3\times10^{-6}$ throughout both examples. These are linear field calculations, not nonlinear halo evolutions.

At $a = 1$, the dust-only contrast is 0.0200000; the mixed contrast is 0.0155804. The dust-only potential stays constant, whereas the mixed potential amplitude falls to 0.779022 of its initial value (Figure 7.1). Density contrast can therefore grow while the potential well becomes shallower as the background expands. That is a consequence of the factor D/a in Eq. (7.12), not inconsistent behavior of the two quantities.

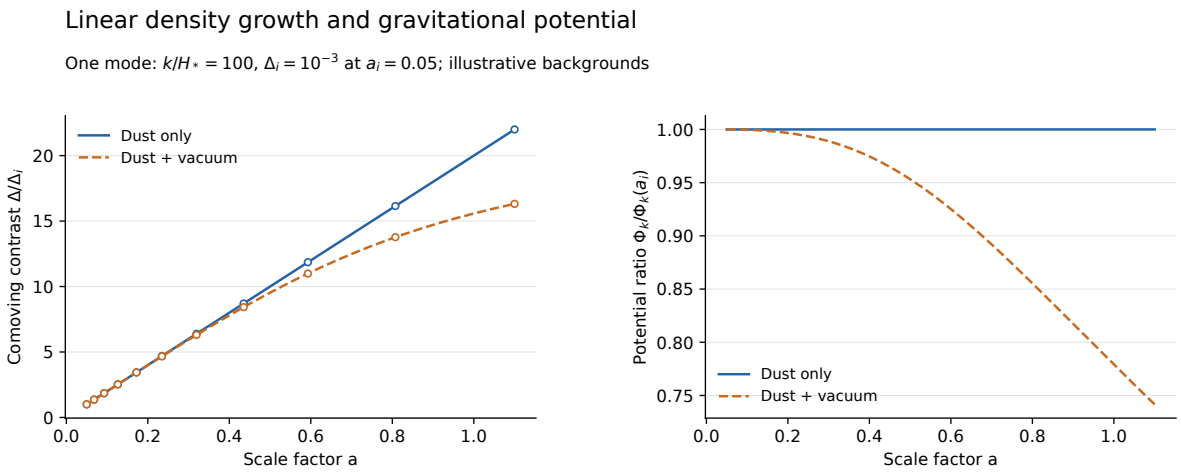


Figure 7.1: Growth and potential of the specified linear mode on two coupled backgrounds. Lines follow Eq. (7.10); open markers reconstruct the contrast from an independent integration of Eq. (7.7) and the momentum constraint. The right panel shows potential relative to its own initial value, not equal initial absolute potentials across the two backgrounds. Both initial density contrasts are 10^{-3}, while their background dust densities differ.

The growth and fluid systems were integrated separately with an eighth-order explicit Runge–Kutta method, using relative/absolute tolerances $(10^{-9}, 10^{-12})$ and $(10^{-11}, 10^{-14})$, a maximum q-step of 0.02, and 401/801 output points. The unused energy constraint was monitored. Growth was also compared with quadrature of Eq. (7.11) at 41 points per background. A separate saved-data audit checked its equivalent hypergeometric form $D = a\,{}_2F_1(1/3, 1; 11/6; -\Omega_{v*}a^3/\Omega_{d*})$, as well as finite-difference derivatives and the constraints. All comparisons met the specified 10^{-6} relative tolerance; these are implementation checks, not rigorous enclosures of nonlinear approximation error.

7.5 Light-travel delay from the same evolving mode

The metric now supplies a propagation calculation rather than leaving the density-to-timing relationship implicit. Light is treated as a test signal with negligible backreaction on the source. From the null condition in Eq. (7.6), the first-order coordinate travel time along a background path with comoving length element $\mathrm{d}\ell$ satisfies $\mathrm{d}\eta = [1 - (\Psi + \Phi)]\mathrm{d}\ell$. Thus

$$\delta\eta = -\int_\Gamma (\Psi + \Phi)\,\mathrm{d}\ell = -2\int_\Gamma \Phi\,\mathrm{d}\ell. \tag{7.13}$$

This is the weak-field gravitational propagation contribution underlying Shapiro delay [10, 11]. Both the temporal and spatial parts of the metric enter. A locally measured null speed remains $c = 1$; the coordinate timing change is not a change of the local light speed.

For a definite example, the ray crosses the positive-contrast half of the mode:

$$x(\ell) = \ell - \frac{\pi}{2k}, \qquad 0 \le \ell \le L = \frac{\pi}{k}, \qquad \frac{\mathrm{d}a}{\mathrm{d}\ell} = a^2 H(a), \qquad a(0) = 1. \tag{7.14}$$

There is no transverse gradient along this planar ray. Equation (7.13) is evaluated on the background null path, retaining the changing potential. The comparison fixes the same coordinate endpoints and emission conformal time in each background and its perturbed version. The endpoint potentials vanish. An observer held at the receiver coordinate measures an additional proper interval

$$\delta\tau_R = a_R\,\delta\eta \quad \text{to first order}, \tag{7.15}$$

where a_R is the background scale factor at arrival. This specified clock comparison, rather than an absolute coordinate delay without a reference, gives the timing its physical meaning. It is not an observed astronomical delay.

Holding the potential at its emission value would instead give

$$\delta\eta_{\mathrm{fr}} = -\frac{4\Phi_k(1)}{k}. \tag{7.16}$$

This frozen reference is exact for the dust-only growing mode at linear order. In the mixed example the evolving calculation is 0.7747% smaller, because the potential continues to weaken during transit. Table 7.1 reports both cases, and Figure 7.2 connects the spatial mode with the accumulated delay. The absolute delays differ because both the background dust density and its growth differ; the table does not isolate vacuum as the only changing input.

Evolving gravitational potentials also produce integrated Sachs–Wolfe frequency shifts [12]; the observable calculated here, however, is the change in light-travel time relative to the frozen-potential reference. The 0.7747% timing difference is not a calculated photon-frequency shift.

Quantity	Dust only	Dust plus vacuum
$\Delta(1)/\Delta_i$	20.0000	15.5804
$\Phi_k(1)/\Phi_k(a_i)$	1.000000	0.779022
Background arrival a_R	1.031663	1.032203
Conformal delay $H_*\delta\eta$	1.200000×10^{-7}	2.782754×10^{-8}
Receiver interval $H_*\delta\tau_R$	1.237995×10^{-7}	2.872367×10^{-8}
Evolving/frozen delay	1.000000	0.992253

Table 7.1: Dimensionless linear-response results for the specified mode and endpoints. No value of H_* in seconds or measured cosmological density is assumed.

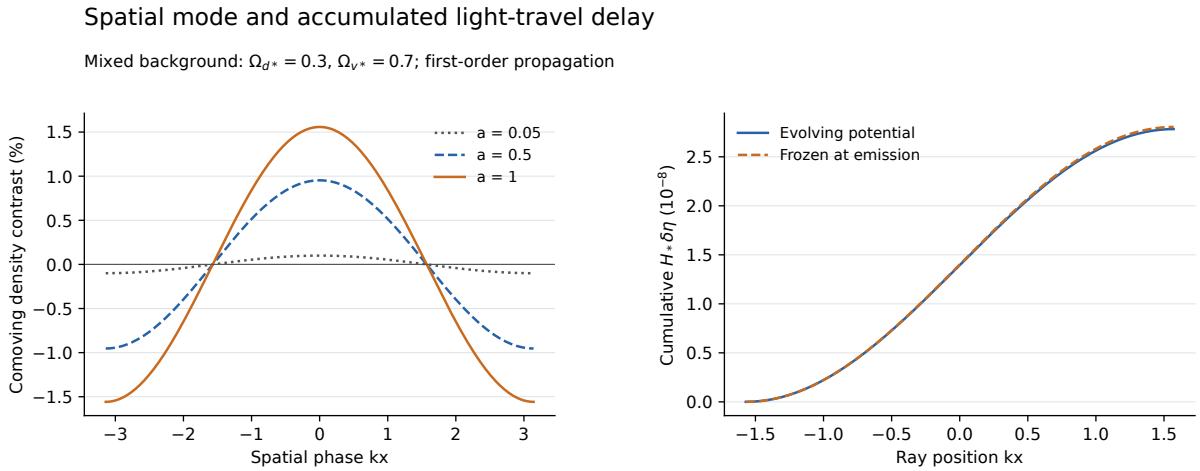


Figure 7.2: Left: the mixed model's comoving density contrast at three scale factors over a full spatial period. Negative contrast denotes density below the positive background, not negative total density. Right: the first-order delay accumulated across its positive-contrast half-period, retaining potential evolution and comparing with the frozen-emission approximation. The mode and timing are calculated consequences of the supplied candidate, not fitted observations.

The ray background and accumulated delay were integrated at both tolerance settings with 401 output points and a maximum path step $L/100$. Independent quadrature and a saved-table Simpson check agreed with the delay to better than 3×10^{-11} relative difference. Zero amplitude gives zero shift, and reversing the mode reverses the first-order shift. Numerical accuracy here concerns the stated linear problem; it does not remove the omitted higher-order density and metric effects.

7.6 Result and significance for the interpretation

The completed addition supplies a specific chain: the candidate stress and conservation laws determine a coupled background; their linear constraints and evolution determine a density mode and its metric potential; that same potential determines a clock-referenced light-travel effect. Within this example, a gravitationally realized dust contribution can support growing density variations and nonzero light-travel delay while keeping locally measured c unchanged. The dust-plus-vacuum option is therefore a mathematically specified candidate, not merely the phrase "space has density."

The comparison also locates the freedom that remains. The density coefficient, initial spatial mode, and geometric data were supplied; they were not derived from an intrinsic property of space or from a Yang–Mills gap. For the specified dust component with zero pressure, zero sound speed, and no anisotropic stress, and the same constant vacuum component, agreement with cold dark matter at linear order follows from the governing equations and matched initial/geometric data [5, 7]. The numerical agreement checks their implementation rather than independently establishing this equivalence. The interpretation of origin is consequently not selected by this agreement. Potential distinctions concern the nonlinear regime, including shell crossing and caustics of the single-flow description [6], or an explicitly modified sound-speed model [7, Sec. 8]; the latter changes the model and is not part of the present calculation. A distinct spatial distribution or dynamical relation could change the comparison, but none is inferred from the present calculation.

This provides a concrete option for a scoped density proposal in the accompanying interpretation of relativity, without choosing it on that interpretation's behalf. The author's explanation that time exists because of space concerns why relativistic relationships should hold; it is interpretive context, not a separate new effect that this calculation was designed to prove. The present mathematical result concerns a candidate density realization and its gravitational consequences.

8 Conclusion

The three descriptions are linked by a common energy balance, but pressure determines how that link operates. In a pressureless description, conserved massive populations and mimetic matter have the same inverse-volume density and the same effective stress tensor when $C = \sum_i m_i N_i$ and their four-velocities agree. The two-species calculation makes the approach to this description quantitative: at its coldest tested state, pressure remains approximately 1.23% of total energy density and total energy remains approximately 1.90% above rest energy.

Vacuum stress joins the same conservation framework with a different pressure. Separate conservation holds its density constant, whereas matter-like density dilutes as J^{-1}. On the selected reference volume their initially equal densities become 1 and 1/169. Imposing the matter-like curve while retaining vacuum-form pressure requires the explicit outflow $Q = -3H\rho_v$, whose integration reproduces the target within the stated numerical tolerance. Thus a shared density history requires more than a shared mass or initial density: it requires compatible pressure and exchange.

The completed comparison makes an effective matter correspondence and its limits explicit. Equal density alone does not imply equal acceleration, clustering, or origin. The mass-gap question supplied a motivation for examining massive states, but the supplied spectrum does not determine their populations or an intrinsic gravitational density of space. For the author's interpretation, the outcome is therefore specific: established mathematics permits a dust-level match under stated assumptions, but neither that match nor its numerical verification identifies dark matter with space itself.

The focused density realization in Section 7 extends this comparison beyond homogeneous matching. It completes the candidate stress with covariant conservation and gravitational coupling, calculates a linear mode on dust-only and mixed backgrounds, and propagates light through the resulting potential. At the reference epoch the mixed mode has grown by a factor 15.5804 while its potential has weakened to 0.779022 of its initial amplitude. Its nonzero light-travel delay is obtained without altering locally measured c, and including potential evolution reduces the frozen-reference delay by approximately 0.775%.

The resulting density-to-metric-to-timing chain gives a concrete effective realization for consideration in the accompanying interpretation of relativity. It agrees with established dust-plus-vacuum gravitational behavior under the same specified inputs, while leaving the proposed intrinsic origin of that density undecided. That distinction is the useful outcome: the candidate's mathematical consequences are explicit, and the interpretation need not be mistaken for a new observational distinction already established by those consequences.

Preparation and Provenance Note

The author supplied the motivating interpretation of an intrinsic gravitational contribution of space, selected and directed the three-way investigation, and requested that the mathematical connections be evaluated before drawing conclusions about that interpretation. The sequence of calculations and the scope of this first draft follow that direction. The author is responsible for the interpretation and for deciding which claims to advance in a final manuscript.

OpenAI Codex assisted with source examination, equation development, implementation and auditing of the calculations, figure preparation, and assembly of this first draft. Established source results, supplied inputs, algebraic consequences, and numerical verification are distinguished in the text.

At the author's direction, GPT 6 added the focused density realization and its bounded perturbation and propagation calculations in Section 7. The author requested the two distinctions guiding that comparison and retained the decision about its adoption in the accompanying interpretation. The original three-way calculation records were preserved. The author's time–space explanation remains interpretive context rather than a separate new-effect claim of the added calculation.

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